INCH-POUND

MIL-R-0043024E(GL)
13 June 1991
USED IN LIEU OF
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25 July 1977

MILITARY SPECIFICATION

REFRIGERATOR, MECHANICAL, COMMERCIAL, PORTABLE, REACH-IN, 70 CUBIC FOOT (LESS MECHANICAL UNIT)

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-R-43024. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This document covers a portable, reach-in field refrigerator of 70 cubic foot capacity (see 6.1).

1.2 Part or identification number (PIN). The part or identification number (PIN) shall consist of the letter M, followed by the basic number of the specification and the appropriate numbering sequence as specified in 6.5.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the Standardization Documents Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4110

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shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-55057	-	Panels, Wood/Wood Based: Constriction and Decorative
QQ-A-200/8	-	Aluminum Alloy 6061 Bar, Rod, Shapes, Tube and Wire, Extruded
QQ-A-200/9	-	Aluminum Alloy 6063 Bar, Rod, Shapes, Tube and Wire, Extruded
QQ-A-200/16	-	Aluminum Alloy 6061, Structural Shapes, Extruded
QQ-A-250/2	-	Aluminum Alloy 3003, Plate and Sheet
QQ-Z-363	-	Zinc-Base Alloy; Die Castings
TT-L-58	-	Lacquer: Spraying, Clear and Pigmented for Interior Use Cellulose Nitrate (General Interior Use)
TT-S-176	-	Sealer, Surface, Varnish Type, Floor, Wood and Cork
TT-V-121	-	Varnish, Spar, Water Resisting
MMM-A-121	-	Adhesive, Bonding Vulcanized Synthetic Rubber to Steel
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-T-60	-	Tape: Packaging, Waterproof

MILITARY

MIL-P-116	-	Preservation Method of
MIL-P-514	-	Plates, Identification, Instruction and Marking, Blank
MIL-T-704	-	Treatment and Painting of Materiel
MIL-S-40144	-	Shelving, Storage, Stationary and Mobile, Food Service
MIL-C-46168	-	Coating, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-S-46163	-	Sealing, lubricating , and Wicking Compounds: Thread-Locking, Anaerobic, Single - Component

STANDARDS

FEDERAL

FED-STD-H28	-	Screw Thread Standards for Federal Services

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

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potential of 500 volts ... Wiresiang ... without flashover or insulation breakdown, a ...

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of US Military Property
MS 35647-5	-	Padlocks, Key Operated
MS 27211-2	-	Thermometer, Indicating, Capillary Tube and Bulb, 2 1/2 Inch Dial Size

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

UNITED STATES DEPARTMENT OF COMMERCE

Product Standard PS-1 - For Construction and Industrial Plywood

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

5-13-2346	-	Refrigerator, Field, Portable, Reach-in (70 Cu Ft) Assembly Complete
5-13-2347	-	Section A-A
5-13-2348	-	Section B-B
5-13-2349	-	Frame Assembly
5-13-2350	-	Sub Floor and Skid Assembly
5-13-2351	-	Lifting Loop Assembly and Skid Details
5-13-2352	-	Panel Assembly
5-13-2353	-	Floor, Floor Rack and Breaker Strip Detail
5-13-2354	-	Door Assemblies
10-1-453	-	Packing Hood, Overseas Shipment for Refrigerator Mechanical

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UX, Natick, MA 01760-5017.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 633 - Electrodeposited Coating of Zinc on Iron and Steel
D 2740 - Tubing, Plastic, Polyvinyl Chloride (PVC)
D 3951 - Standard Practice for Commercial Packaging
D 3953 - Steel, Flat and Seals, Strapping

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Standard No. 70 - The National Electric Code

(Application for copies should be obtained from the National Fire Protection Association, Batterymarch Park, Quincy, MA 02169.)

AMERICAN NATIONAL STANDARD / AMERICAN HARDBOARD ASSOCIATION ANSI/AHA

A 135.4 - Basic Hardboard

(Application for copies should be obtained from the American Hardboard Association, 520 N. Hicks Road, Palatine, Illinois 60067)

UNDERWRITERS LABORATORIES INC. (UL)

UL 498 - Receptacles, Attachment Plugs
UL 1571 - Incandescent Lighting Fixtures
UL 1581 - Wires and Cables, and Flexible Cords, Electrical

(Application for copies should be obtained from Underwriters Laboratories, INC., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

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3.2 Codes and standards. Refrigerator electrical components shall comply with the requirements of Underwriter Laboratories Standards No. 1581, 1571 and 498 and National Fire Protection Association (NFPA) Standard No. 70 (see 4.4.7, 4.4.7.1 and 4.4.7.2).

3.3 Materials and components. Material and components shall be as specified herin and on the applicable drawings and shall be of the quality normally used by the manufacturer for refrigerators, provided the completed end item complies with all the provisions of this document (see 6.4). The material shall have no adverse effect on the health of personnel when used for its intended purpose. Cadmium, antimony, bismuth, lead, or other toxic materials shall not be used on any hardware or components in the interior of the the refrigerator. Commercial tolerances shall apply to all metal thicknesses and diameters unless otherwise specified hereinafter or on drawings. Components offered as being equivalent to components specified hereinafter and on the reference drawings as a specific manufacturer's part number or equal shall be functionally equal, of equal or better quality, and shall have the manufacturer's part number identified. The incorporation and inclusion of such a component in the design of the specified end product shall not require modification or change to any other components, and shall not reduce ease of maintenance to it or any other components unless such modification or change is specifically approved by the contracting officer.

3.3.1 Insulation.

3.3.1.1 Foamed-in-place insulation. Foamed-in-place insulation shall be polyurethane material and have a manufacturer's rated "K" factor of not more than 0.15. Insulation shall have a free rise density of not less than 1.7 pounds per cubic foot when tested as specified in 4.5.2 or an in-place density of not less than 2 pounds per cubic foot when tested as specified in 4.5.2.1. The foamed-in-place insulation shall be dimensionally stable at a temperature of 150 degrees F when tested as specified in 4.5.1.

3.3.2 Zinc-alloy die castings. When die cast zinc-alloy is used, the zinc base alloy shall conform to alloy AG40A of QQ-Z-363.

3.3.3 Aluminum alloy.

3.3.3.1 Extruded shapes. Extruded aluminum alloy shapes shall conform to alloy 6061-T6510 of QQ-A-200/8 or alloy 6063-T6 of QQ-A-200/9.

3.3.3.2 Rolled or drawn shapes. Aluminum alloy for rolled or drawn shapes shall conform to alloy 6061-T6 of QQ-A-200/16.

3.3.3.3 Sheet. Aluminum alloy sheet shall conform to alloy 3003, temper H 14 or H 16 of QQ-A-250/2 as specified on the drawings.

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3.3.4 Gasket. The door sealing gasket shall conform to drawing 5-13-2354. Gasket shall be adhered in place using adhesive conforming to MMM-A-121. All joints and ends shall be sealed.

3.3.5 Fasteners. Unless otherwise specified herein, fastener hardware shall have the contractor's standard electrodeposited finish and threads shall be class 2 fit and conform to FED-STD-H28.

3.3.6 Building board. Material for the front and side assembly and refrigeration unit opening breaker strips shall conform to class 2, (S1S) of ANSI/AHA, A135.4, surface shall be unfinished plainboard.

3.3.7 Plastic tubing. The plastic tubing, as used to shield the door latch plunger from insulation as shown on Drawing 5-13-2354 shall conform to ASTM D 2740.

3.3.8 Metal surface priming and coating.

3.3.8.1 Primer coat for metal surfaces. Primer for ferrous and non-ferrous metal surfaces shall conform to type F or G of MIL-T-704.

3.3.8.2 Finish coat for metal surfaces. The primed exterior metal surfaces shall shall be painted green 383, 34094, type II, III or IV of MIL-C-46168 or green 383 of MIL-C-53039.

3.3.8.3 Finish, panel; interior and floor. The interior panels (items 4-4, 4-5, 4-6 and 4-7 of Natick Drawing 5-13-2352), the inner floor (item 9 of Natick Drawing 5-13-2346 and shown on Natick Drawing 5-13-2353), and inner skin (item 5-2 on Natick Drawing 5-13-2354) are to be clear anodized after forming or welding and conform to type II, class 1 of MIL-A-8625, as specified in MIL-T-704.

3.3.9 Sealer, varnish and lacquer.

3.3.9.1 Hardboard sealer. Hardboard sealer shall conform to class II of TT-S-176.

3.3.9.2 Varnish. Varnish shall conform to TT-V-121. Varnish prepared for spraying shall have not more than 1 pint of varnish makers and painters naphtha (VM&PN) added to each gallon of varnish.

3.3.9.3 Lacquer. Lacquer shall conform to type I, class II of TT-L-58.

3.3.10 Plywood. Plywood shall be Douglas Fir or Southern Pine, exterior type, grade B-C or better, conforming to PS-1.

3.3.11 Shelving. When specified (see 6.2), shelving conforming to MIL-S-40144 as required shall be furnished.

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3.3.12 Floor racks. Fiberglass or plastic floor racks as shown on drawing 5-13-2353 shall be provided.

3.4 Design. The design of the refrigerator covered by this document shall conform to the requirements herein and to the drawings listed in 2.1.2.

3.4.1 Heat Leakage. The heat leakage of the refrigerator shall not exceed 15 BTU/HR/^{o}F when tested as specified in 4.5.5.

3.4.2 Rainfall test. The refrigerator, with internal light on, shall withstand a rainfall of 3 inches per hour without leaking or showing damage when tested as specified in 4.5.3.

3.5 Construction.

3.5.1 Metal construction. Metal seams shall present a smooth exposed surface. The refrigerator shall not show any damage when tested as specified in 4.5.4, using the lifting rings and with the refrigerator containing an evenly distributed load of 2600 pounds. Welded seams shall be made at inner floor and where metal sheets are notched to accommodate forming.

3.5.2 Breaker strips. The breaker strips shall be constructed of material specified in 3.3.6 and shall be removable with the insulation side treated to prevent adherence to insulation.

3.5.3 Insulation. All space within the walls, door, floor and top shall be completely filled with foam in-place insulation as specified on drawing 5-13-2347. Prior to foaming, the inner and outer liner shall be treated in accordance manufacturer's standard commercial practice to insure permanent bonding of the insulation to the liner material. The insulation shall be installed as to leave no voids or empty pockets.

3.5.4 Doors.

3.5.4.1 Door latch and strike assembly. The reach-in door latch and

5-13-2354. The latch and strike assembly shall be fabricated of high strength zinc cast conforming to 3.3.2, cast or forged brass, or malleable iron with zinc, chromium or nickel finish. When CARC is specified (see 6.2), hardware components shall be supplied with finish compatible for CARC primer and paint application (see 3.3.8). The latch shall have provisions for holding the padlock specified in 3.5.4.3. When a diecast latch is furnished, the handle shall be forged brass or malleable iron and the striker tongue shall be extruded brass or forged steel. The strike shall be so constructed that removing the bolt or screw from the strike housing shall not allow the locked door to open. The door latch shall be provided with an internal release mechanism designed so that the door may be opened from the inside even though the latch may be secured or padlocked on the outside and in so doing no components shall become distorted, dismounted or detached.

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3.5.4.2 Hinges. Hinges for the reach-in door shall be of commercial self-closing type as identified on Drawing 5-13-2354 and shall match the latch of the door in appearance. The hinge shall be not less than 10 inches long. The hinges shall be constructed of zinc die cast conforming to 3.3.2, steel, forged brass, or malleable iron. The hinge shall be finished with zinc, chromium or nickel. When CARC is specified (see 6.2), hardware components shall be supplied with finish compatible for CARC primer and paint application (see 3.3.8). Lubrication shall be accomplished through the use of self-lubricating nylon cam and cam insert. Shims shall be provided behind door hardware to compensate for gasket set.

3.5.4.3 Padlocks. A padlock conforming to MS 35647-5 shall be supplied. The padlock shall be securely attached to the reach-in door by chain as provided.

3.5.4.4 Two point latch. The reach-in door shall be provided with 2- two point latches as shown on Drawing 5-13-2354.

3.5.5 Hardware, latch, strike and hinge. Retention hardware for the door latch, strike and hinge shall be secured in place with sealing compound, type II, grade N conforming to MIL-S-46163.

3.5.6 Electrical. Unless otherwise specified (see 6.2), all electrical devices and components shall be installed as shown on the drawing and be capable of operating on a nominal 115 volt, 60 hertz, single phase power system. Electric wire shall be nonmetallic-sheathed cable conforming to type NMC or UF of the National Electric Code. The cable shall contain two conductors and a ground wire. The conductor size shall be No. 14 AWG. The cable shall conform to Underwriters Laboratories Standard No. 1581.

3.5.6.1 Lighting Fixture. An interior, vapor-proof light fixture, for horizontal surface mounting with a rotary switch mounted on its base, shall be supplied and installed as shown on drawing 5-13-2346. The interior light fixture shall be located on the walk-in-door panel on the latch side of the door with the horizontal center line of the fixture approximately in line with the top of the door. The vapor-proof lamp fixture shall be equipped with a shatterproof globe of the polycarbonate screw type and shall be complete with a 60-watt, 120-volt frosted, incandescent bulb having a medium screw base. The light fixture shall conform to Underwriters Laboratories Standard No. 1571.

3.5.6.2 Plug and receptacle. The plug and receptacle conforming to Underwriters Laboratories Standard No. 498 shall be provided and shall be weather-tight, designed for 3-wire, 2-pole grounded service and shall be rated at not less than 15 amperes. The receptacle shall be of the recessed male, locking type. A cap or flap cover shall be provided to seal the receptacle housing when not in use and shall be secured to the housing to prevent loss. The plug shall contain a sealing ring to form a weather tight joint when the plug is inserted in the receptacle.

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3.5.6.3 Grounding rod. For Navy procurements (see 6.2), provisions shall be made to permanently ground and bond all exposed metal sheathing of all refrigerator walls, ceiling, floor, and door, both inside and outside of the refrigerator to a grounding rod.

3.5.7 Thermometers. A thermometer shall be provided conforming to MS 27211-2 and shall be mounted as shown on Drawing 5-13-2346.

3.5.8 Finish. Unless otherwise specified (see 6.2), all external surfaces and components shall be finished.

3.5.8.1 Cleaning, conditioning and surface preparation. All exposed outer surfaces shall be thoroughly cleaned, conditioned and prepared for painting in conformance with MIL-T-704.

3.5.8.1.2 Primer coat. All metal surfaces prepared for painting shall receive a coat of primer in accordance with type F or G of MIL-T-704 (see 3.3.8.1), after having thoroughly dried from the cleaning and conditioning process (see 3.5.8.1).

3.5.8.1.3 Finish coat. Unless otherwise specified (see 6.2), the surfaces shall receive a coat of the finish paint specified in 3.5.8.2, after the primer has cured to the proper degree for painting.

3.5.8.2 Hardboard surfaces. The hardboard surfaces of the door jambs and reach-in door and the refrigeration unit opening shall be given one coat of sealer as specified in 3.3.9.1 followed by two coats of varnish specified in 3.3.9.2 or lacquer specified in 3.3.9.3.

3.5.9 Marking.

3.5.9.1 Data name plate. The refrigerator shall have a metal identification plate attached to the outside of the reach-in door as shown on Drawing 5-13-2354. The metal plate shall conform to type III, composition A, class 1 or 2 of MIL-P-514 and shall be permanently attached with screws or rivets. The identification plate shall be marked in accordance with MIL-STD-130 as follows:

1. U.S.
2. Military model number and nomenclature.
3. National stock number (NSN)
4. Serial number.
5. Contract number.
6. Manufacturer's name and address.
7. Model number.
8. CAGE number.
9. Technical manual number.
10. Weight.

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Each name plate shall be placed so as to be readily visible to the operator during normal operation and use and so as not to adversely affect the life or utility of the refrigerator.

3.5.9.2 Data plate. For Marine Corps acquisitions (see 6.2) a data plate shall be attached to each refrigerator. These plates may be obtained from the Commandant of the Marine Corps (Code CS6), Headquarters Marine Corps, Washington D.C. Data applicable to blank spaces 1, 3, 8, 9 and 10 shall be furnished with the plates. The contractor is required to imprint the data in all blank spaces and permanently affix the plate to the refrigerator.

3.5.9.3 CARC paint identification. Following the application of the finish coat specified in 3.5.8.1.3, and when the finish coat has cured to the proper degree, the word "CARC", along with the month and year of coating application shall be stenciled in 1-inch block letters, using lusterless black paint 37030, type III or IV, conforming to MIL-C-46168. The CARC marking shall be located conspicuously near the data name plate (see 3.5.9.1) so as to be plainly visible during normal operation of the unit.

3.5.10 Workmanship.

3.5.10.1 Welding. The surfaces of all parts to be welded shall be free from rust, corrosion, scale, paint, grease or other foreign matter. Parts to be welded shall be cleaned, ground, shaped, and welded in such a manner as to be sound, smooth, free from flashing, show good fusions, and provide strength comparable to that of the connected parts which have not been impaired.

3.5.10.2 Bolts and screws. Bolts and machine screws shall be securely fastened in place and sheet metal screws shall be securely screwed in place and not driven.

3.5.10.3 Zinc-alloy die castings. Each die casting shall be of a uniform quality and condition, free from foreign matter, blow holes, porosity, hard spots, shrinkage defects, cracks, and other injurious defects that may effect the appearance or may adversely effect the serviceability. The castings shall be made smooth by filing and tumbling before the final finish is applied.

3.5.10.4 Sheet metal. Metal used in the fabrication of the container shall be free of kinks and unspecified bends. Forming and shearing shall not cause damage to the metal and shall be done neatly and accurately. Corners shall be square and true and all bends of a major character shall be made with dies or fixtures to insure uniformity of size and shape. Sharp edges, corners and burrs shall be removed.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested for the characteristics specified in 4.5.1, 4.5.3, and 4.5.5.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

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4.4.2 In-process inspection. Inspection of sub-assemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance.

4.4.2.1 In-process examination. Examination shall be made to establish that no deviation is made from the following requirements:

a. Examine hardboard and plywood for type specified (see 3.3.6 and 3.3.10).
b. Breaker strip treatment (see 3.5.2).
c. Insulation is installed with no voids.
d. Where applicable, metal surfaces are properly treated for bonding with foamed insulation.
e. Electrical insulation is in conformance with 3.5.6 and applicable drawings.
f. Finish, cleaning, conditioning, and priming is as specified in 3.5.8.1.
g. Welding is in conformance to 3.5.10.1 and applicable drawings.
h. Rivet holes punched cleanly and accurately.
i. Riveting uniform and installed properly.
j. Hardware, fasteners properly positioned.

4.4.2.2 In-process test of insulation. The density tests of foamed-in-place insulation shall be performed as specified in 4.5.2 or 4.5.2.1. The inspection lot shall be the amount of material used in one day's production. The sample unit shall be a 6-inch cube.

4.4.2.3 In-process dimensional examination. Prior to assembly, components shall be examined for compliance with all specification and drawing dimensional requirements. The inspection lot shall be all like components offered for inspection at one time. The inspection level shall be S-2 with an acceptable quality level (AQL) of 2.5 expressed in terms of defects per hundred units.

4.4.3 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of refrigerators. The sample unit shall be one refrigerator. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major and 6.5 for total (major and minor combined) defects.

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TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Type of finish not as specified		201
	Blistered, peeled, chipped area of no film, runs, wrinkles, or drops	101	
	Color not as specified		202
Construction and workmanship	Hardboard and plywood members not one piece		203
	Sharp seams on inner or outer skin (when applicable)	102	
	Kinks or unspecified bends	103	
	Seams not welded on inner floors	104	
	Joints of notched metal sheet not welded		204
	Corners and angles on inside surface not filled or rounded	105	
	Bolts, screws, defectively installed		205
	Defective castings		206
	Latch and strike assembly not adjustable (when applicable)		207
	Refrigerator door cannot be opened from inside when latch is secured or padlocked	106	
	Components not attached when refrigerator door is opened from the inside		208
	Latch cannot be locked with padlock		209
	Shims not provided behind door hardware		210
Electrical system	Defective pilot light		211
	Plug receptacle not as specified		212
	Plug not furnished		213
	Light fixture not mounted on as specified		214
	Not grounded when specified	107	
	Device capacity rating incorrect		215
Welding	Missing, incomplete fusion	108	
	Rust or corrosion on areas to be welded	109	
	Scale, paint or grease not removed		216
Marking	Missing, incomplete, illegible, not in proper location, not on correct data plate		217

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Padlock	Not provided	110	
	Not attached as specified		218

4.4.4 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end item shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of refrigerators. The sample unit shall be one refrigerator. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred, shall be 4.0.

4.4.5 End item testing. When a first article is not required, the initial production unit shall be tested as specified in 4.5.1, 4.5.3, 4.5.4, and 4.5.5. Any nonconformance shall be cause for rejection of the initial production unit.

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or otherwise defective
Workmanship	Inadequate application of components such as: incomplete taping or plywood not flush against refrigerator (where required).

4.4.7 Certification compliance examination. Certifications, certified test reports, or listing marks for standards, submitted in accordance with 3.2, shall be examined and validated as proof of compliance. Prior to Government approval of first shipment, the contractor shall submit to the contracting officer or his authorized representative for approval, certification that the refrigerators he proposes to furnish, meet the applicable requirements of UL as follows.

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4.4.7.1 UL. Acceptable evidence of meeting the requirements of the Underwriters Laboratories Standards shall be a UL listing mark, or a certified test report from a recognized independent laboratory, acceptable to the Government, indicating that the refrigerators have been tested and conform to the UL Standards No. 498, 1571 and 1581.

4.4.7.2 NFPA. Acceptable evidence of meeting requirements of the National Fire Protection Association (NFPA) shall be the manufacturer's certified statement that the electrical components conform to the applicable requirements of NFPA No. 70.

4.5 Methods of inspection.

4.5.1 Heat stability test. One complete production unit shall be dimensionally measured for wall thickness (include center of ceiling) and placed in an oven or heated room and held at 150 degrees F for 24 hours. After removal of the unit from the oven, any indication of swelling shall require a measurement to determine if the foam had a swell of more than 5 percent to determine conformance to 3.3.1.2. Inability of the insulation to remain stable within 5 percent shall constitute failure of the test.

4.5.2 Free rise density for insulation. Prior to each day's production, one free foam sample of the insulation referenced in 3.3.1.1, a minimum 8-inch cube, shall be taken. From the center of the sample a 6-inch cube shall be cut and weighed. The density shall be determined as follows:

Density (lbs/cubic foot) = Weight of 6-inch cube (lbs) X 8

Inability of sample to conform to density requirements of 3.3.1.1 shall constitute failure of the test.

4.5.2.1 In-place density test for insulation. Prior to each day's production the first refrigerator to be foamed shall be identified, weighed and the weight recorded. The refrigerator shall then be foamed and upon completion, weighed and the weight recorded. Divide the total pounds of foam (weight differential before and after foaming) by the interior volume of the space being insulated to determine conformance with the in-place density requirement of 3.3.1.1. Inability to conform with the in-place density requirement shall constitute failure of this test.

4.5.3 Rainfall test. The door end of the refrigerator shall be sprayed with fresh water at a rate of 3-inches per hour at an average angle of 45 degrees for 1 hour. Following the simulated rainfall, the refrigerator shall be carefully examined to determine compliance with 3.4.2. Leakage or damage shall constitute failure of this test.

4.5.4 Lift test. To determine conformance to the lifting requirements of 3.5.1, the following test shall be performed:

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The interior of the refrigerator shall be loaded internally with an evenly distributed weight of 2600 pounds. The refrigerator shall be lifted approximately 5 feet using the four lifting rings, and then lowered to the ground. This shall be repeated two additional times and then the refrigerator shall be examined for any damage. Any damage to the refrigerator shall constitute failure of this test.

4.5.5 Heat leakage test. The refrigerator used for the heat leakage test shall have an insulated plug in the refrigeration unit opening. A heat leakage test shall be conducted with the refrigerator placed in a controlled temperature ambient and a monitored heat input supplied to the refrigerator to determine compliance with 3.4.1. All calculations, using the formula below, and all recorded data on time, temperature and heat input shall be recorded.

$$\text{Heat leakage} = \frac{\text{Total watt-hours input X Btu/watt-hour}}{\text{Length of test (hours) X temperature difference (F)}}$$

The temperature difference shall not be less than 50 degrees F. The interior of the refrigerator temperature shall not exceed 140 degrees F. The ambient temperature shall be the average of eight readings taken at points located 6-inches from each outside corner of the box. The internal, or box temperature, shall be the average of eight readings taken at points located 6-inches from the inner walls a each corner of the box. Eight ambient and box temperature readings shall be taken at consecutive one-half hour intervals with ambient temperature variations not more than $\pm$ 2 degrees F and box temperature variations not more than $\pm$ 1 degree F. At least one interior and one exterior temperature reading shall be taken on a recording instrument. The refrigerator shall be no closer than 12-inches from any wall or ceiling. The inability to meet the heat leakage requirement of 3.4.1 shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. The locking assembly on the door of the refrigerator shall be removed and together with the hardware, accessories, padlock, and keys shall be unit packed conforming to method III of MIL-P-116 in a snug-fitting fiberboard box conforming to style RSC, class domestic of PPP-B-636. The packed components shall be secured within the refrigerator in such a manner as to prevent movement or damage while in transit.

5.1.2 Commercial. The refrigerator shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or commercial as specified (see 6.2).

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5.2.1 Level A packing. Each refrigerator, preserved as specified in 5.1, shall be provided with a snug-fitting wood cleated, plywood hood as shown on Drawing 10-1-453, with material and nailing requirements conforming to overseas type PPP-B-601, and the following: The cleats for the top of the hood shall be placed on the the inside of the hood, and the top of each refrigerator shall be either covered with a sheet of minimum 40-pound kraft paper or minimum 0.002-inch thick polyethylene. Each hood shall be strapped to the refrigerator by means of four straps, two encircling the girth of the hood and passing through the openings in the skid of the refrigerator closest to the ends and two encircling lengthwise over the hood and under the refrigerator and adjacent to the skids of the refrigerator. Strapping shall be minimum 3/4 by 0.023-inch steel conforming to ASTM D 3953.

5.2.2 Level B packing. Each refrigerator shall have the plug panel mounted in the unit opening and secured in place with hardware furnished. The receptacle cap shall be screwed on finger tight. The thermometer and the pilot light assembly shall be covered and protected with plywood fitted to bear flush against the refrigerator. The plywood shall be secured in place by means of straps of tape covering all sides and extending onto the surfaces of the refrigerator. The tape shall be minimum 2-inch width conforming to type III or type IV, class 1 of PPP-T-60. The plywood shall be not more than 1/4-inch thick and shall conform to type A of A-A-55057. The plywood shall extend a minimum of 1/2-inch beyond the edges of the thermometer or pilot light on all sides. The door shall be closed and locked with a padlock when furnished and the keys taped to the padlock with tape specified herein; or shall be wired shut with minimum 0.072-inch diameter annealed wire.

5.2.3 Commercial packing. Each refrigerator, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The refrigerator is intended for use with the refrigeration units of MIL-R-43031 and is intended to be used to store temperature sensitive food items and medical supplies at temperatures ranging from 0 to 35 degrees F under adverse climatic and operational conditions in the field without any further protection from the elements.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

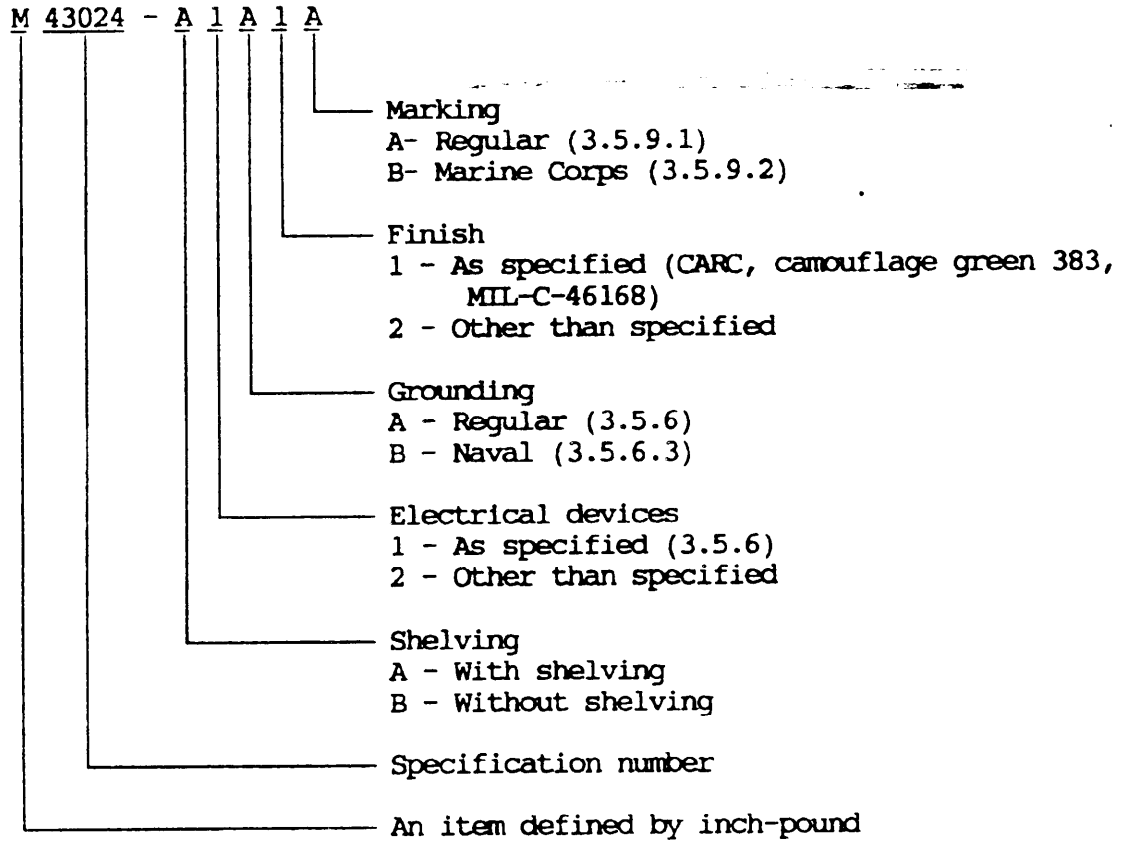
a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. When shelving is required (see 3.3.11).
e. When tiedown assemblies are required (see 3.5.5).
f. When electrical devices are to be installed other than specified (see 3.5.6).
g. When special ground for Navy is required (see 3.5.6.3).
h. When exterior finish is other than specified (see 3.5.8.1.3).
i. When procurement is for Marine Corps (see 3.5.9.2).
j. Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).
k. When overseas type packing hood is required (see 5.2.1).

6.3 First Article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.3).

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6.5 Part number. The refrigerator shall be identified by a definitive part number (see 1.2) as shown in the following example:



6.6 Subject term (key word) listing.

Kitchen equipment
Refrigerator
Storage, food

6.7 Changes from previous issue. Asterisks are not used in this revision to identify the changes with respect to the previous issue, due to the extensiveness of the changes.

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Custodians:

Army - GL
Navy - YD
Air Force - 99

Review activities:

Army - MD, TS
Air Force - 82
DLA - GS

User activities:

Army - CE
Navy - MC

Preparing activity:

Army - GL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-0043024E(GL)	1991 June 13

3. DOCUMENT TITLE
REFRIGERATOR, MECHANICAL, COMMERCIAL, PORTABLE, REACH-IN, 70 CUBIC FOOT (LESS MECHANICAL UNIT)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
U.S. Army Natick RD&E Center	(1) Commercial: 508-651-4085 (2) AUTOVON: 256-4085
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-WEE Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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